

Project SENTINEL: Mechanistically Interpretable Steering of Foundation Models for De Novo Protein Binder Design

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Motivation

Engineered protein biosensors offer real-time disease detection, but generative binder-design models remain black boxes: candidate binders are produced without characterizing the internal reasoning that separates a high-affinity binder from a poor one. This opacity limits rational design improvement, failure-mode diagnosis, and cross-disease adaptation.

Approach

We train sparse autoencoders (SAEs) on residue-level activations of ESM-C (a state-of-the-art protein language model), extracted from Boltz-designed binder candidates, to recover human-readable features underlying binder representations. As proof-of-concept, we validate this pipeline end-to-end on Vilip-1, a blood-circulating biomarker of acute neurological injury.

Results

- Diversifying SAE training data – folding in real, evolutionarily distinct sequences alongside synthetic designs, and mixing in binder-design data from additional targets (UCH-L1, B-FABP) – substantially improved generalization to real proteins beyond synthetic designs alone.
- Independently trained SAE dictionaries – including one incorporating binder-target co-attention context via ESM-C’s native chain-break token – converge on the same residues in real proteins for a subset of examined features, corroborated by InterPro domain annotations and calibrated LLM-assisted labeling.
- This convergence holds up against the obvious confound (shared random initialization across runs): the agreeing dictionaries mostly rely on different learned feature indices to detect the same real residues, evidence of genuine independent convergence rather than an artifact of training setup.

Figure 1 (TODO). Cross-dictionary consistency / InterPro corroboration figure – placeholder.

Ongoing & Future Work

This interpretability foundation is the necessary precursor to our ultimate goal: steering generative design toward high-affinity binding features, analogous to concept-guided steering in text-to-image diffusion models, rather than blind latent-space sampling. We are extending this pipeline to UCH-L1, S100B, and B-FABP – SENTINEL’s three clinical targets – and evaluating whether SAE features predict binding-quality metrics (binding confidence, ipTM, ipSAE) directly, toward establishing mechanistic interpretability as a universal accelerant for biosensor design.